

A single-parameter geometric model of contextual overlap in the Gender–Science IAT

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Abstract

Implicit Association Tests (IATs) produce robust reaction-time asymmetries between congruent and incongruent pairings, but standard analyses model these effects with additive learning and noise terms. We test a geometric interference account in which contextual overlap is summarized by a single interference angle θ . For fixed θ , participant-level coefficients have closed-form OLS solutions, and profiling θ over a dense grid yields an analytic posterior, $p(\theta) \propto \text{RSS}(\theta)^{-n/2}$. Using public trial-level logs from the Gender–Science IAT ($N = 141,329$), we obtain a sharply concentrated posterior: mean 17.29° , SD 0.48° , 94% HDI $[16.50^\circ, 18.25^\circ]$. This solution is stable across binning schemes, bootstrap resamples, and corrected D-like congruency quintiles (17.00° – 17.75° despite D-like means from -0.32 to 0.83). A matched 2019 Sexuality IAT ($N = 207,366$) remains low-angle (mean 19.12°). An independent life-satisfaction IAT initially produced a near-orthogonal estimate when item identifiers were used as the within-block axis, but a stimulus-order audit showed that this was driven by item-family/order structure: using actual display order returned a low-angle MAP of 17.75° . Additive baselines match local RMSE but leave more structured residual drift and worse held-out distributional fit. Recovery analyses show that low-angle inference requires large-scale aggregation; null and sensitivity analyses separate descriptive stability from preprocessing artifacts. The most stable signal in these data is task-geometric and context-dependent rather than participant-specific.

Keywords: Implicit Association Test, quantum cognition, interference angle, contextual overlap, geometric model, profile posterior

1 Introduction

The Implicit Association Test (IAT) remains the most widely adopted behavioral paradigm for studying implicit social cognition, automatic evaluation, and stereotype activation (Greenwald et al., 1998, 2003; Kurdi et al., 2021). Despite its impact, standard analyses have typically decomposed reaction times into additive contributions from congruency, learning, and noise, relying on linear mixed-effects models and, in some cases, sequential-sampling and diffusion formulations (Klauer et al., 2007; Ratcliff and McKoon, 2016). These approaches provide detailed accounts of trial-level evidence accumulation but are grounded in classical probabilistic assumptions that treat processing channels as independent and errors as Gaussian. As a result, they do not explicitly represent contextual interference arising from overlapping evaluative representations, a phenomenon central to nonclassical and quantum-inspired models of cognition (Townsend and Wenger, 2004; Busemeyer et al., 2006).

A further motivation for shifting the level of analysis comes from ongoing psychometric debate about what can be reliably inferred from IAT performance. Critics have argued that behavioral interference measures do not uniquely identify latent cognitive constructs at the individual level, even when fitted with sophisticated process models (Schimmack, 2021). At the same time, responses to this critique emphasize that the validity of the IAT depends on the construct, task structure, and level of aggregation and caution against both overinterpretation and wholesale dismissal of implicit measures (Kurdi et al., 2021). Recent methodological work strengthens that caution by showing that IAT measurement properties depend on stimulus validation, response modality, and the particular indirect-measure architecture used (Hogenboom et al., 2024; Bar-Anan and Hershman, 2024; Axt et al., 2024).

Taken together, this literature suggests that the strongest and most defensible inferences concern task-level or population-level structure rather than individual latent parameters. Accordingly, the present approach targets population-level contextual geometry, estimating a global interference structure that is identifiable under aggregation without claiming access to individual implicit attitudes.

Different from classical approaches, quantum cognition frameworks treat cognitive states as complex probability amplitudes evolving in Hilbert space (Busemeyer and Bruza, 2012; Pothos and Busemeyer, 2013; Pothos et al., 2022; Huang et al., 2025). Contextual change is represented as a basis rotation between incompatible observables, with interference terms producing empirically verified deviations from classical probability, such as order effects (Trueblood and Busemeyer, 2011; Wang et al., 2014) and decision inertia (Yearsley and Pothos, 2016). The formal analogy between these models and wave mechanics suggests that adaptation dynamics in evaluative tasks can be characterized geometrically rather than additively, with latent “angles” quantifying the overlap between incompatible mental representations (Blutner and beim Graben, 2016; Huang et al., 2025).

Here we introduce a geometric model of implicit adaptation in which contextual overlap is summarized by a single interference angle, θ . The model does not treat θ as an individual attitude parameter or a learning rate. It treats θ as a population-level descriptor of overlap between task contexts in the Gender–Science IAT. For any fixed θ , participant-specific coefficients are estimated by closed-form ordinary least squares. Profiling θ over a dense grid yields an analytic marginal posterior, $p(\theta) \propto \text{RSS}(\theta)^{-n/2}$, without high-dimensional sampling. Highest-density intervals and posterior summaries were computed directly from the discretized profiled posterior; information criteria are reported only as secondary diagnostics.

We ask six questions. Can a single angular parameter summarize adaptation dynamics across 141,329 participants? Is that parameter numerically stable and identifiable at the population level? Does it survive stratification by a conventional congruency metric? Does

it replicate in a matched second public IAT domain? How does the geometric model compare with additive baselines under leak-free temporal hold-out? And how sensitive is the estimate to binning resolution and stimulus-order encoding?

The aim is inference, not short-horizon prediction. We use predictive performance as a diagnostic, but the main target is a stable task-level descriptor of contextual overlap in the IAT.

2 Methods

2.1 Data and preprocessing

We merged trial-level logs `iat.txt`, `iat2.txt`, and `iat.0003.txt` for task `scienceiat` (Project Implicit, demo release). Critical blocks (3, 4, 6, 7) were retained. Within each participant and block, trial index was normalized to $x \in [0, 1]$. Latencies were aggregated to $N_{\text{bins}} \in \{4, 6, 8\}$ quantile bins (main analysis uses 6). Participants with < 3 populated bins were excluded, yielding curves $\{(x_t, y_t)\}_{t=1}^{T_i}$ with $T_i \in [3, 8]$.

Leak-free standardization. For hold-out experiments, z-scoring is computed *per participant from training bins only* and applied to both train and test bins. This prevents outcome leakage into the standardization step. For the posterior over θ (which is not a predictive split), z-scoring uses all bins to align scales with a Gaussian observation model.

2.2 Model

We model standardized mean latency as

$$y_i(x) = a_i + b_i \cos(\omega x) + c_i \sin(\omega x) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad \omega = \frac{\pi}{180} \theta. \quad (1)$$

For any fixed θ , the coefficients (a_i, b_i, c_i) have the closed-form OLS solution $\hat{\beta}_i(\theta) = (X_i^\top X_i)^{-1} X_i^\top y_i$ with design columns $1, \cos(\omega x), \sin(\omega x)$.

2.3 Identifiability and inference

The functions $\{1, \cos(\omega x), \sin(\omega x)\}$ are linearly independent on any interval for $\omega \neq 0$, so $X_i^\top X_i$ is full rank when at least three distinct x values are available. A small ridge term (10^{-5}) was added to stabilize inversion in near-degenerate cases.

We assess identifiability at the population level rather than the participant level. With only $T_i \in [3, 8]$ binned observations per participant, individual likelihoods are too sharp and too unstable to support strong interpretation. Pooling across participants yields a global interference angle that is estimable even when participant-level fits are noisy.

Stacking all participants, we compute

$$\text{RSS}(\theta) = \sum_i \|y_i - X_i \hat{\beta}_i(\theta)\|_2^2,$$

and form the profile posterior

$$p(\theta) \propto \text{RSS}(\theta)^{-n/2},$$

corresponding to a Gaussian model with σ integrated out under a reference prior. We profile θ on a grid $[0^\circ, 180^\circ]$ in 0.25° steps and refine near the mode. Posterior summaries and HDIs were computed directly from the discretized profiled posterior.

2.4 Predictive model comparison

We evaluate five comparators under *temporal* hold-out per participant (highest- $x \approx 20\%$ of bins as test; remainder as train, requiring $\geq$ regressors): (i) full interference; (ii) cosine-only; (iii) sine-only; (iv) polynomial baseline $a + d_1 x + d_2 x^2$; (v) exponential $a + d e^{-kx}$ (grid on k); and (vi) power law $a + d x^p$ (grid on p). All fits use closed-form OLS with ridge 10^{-5} . Performance is measured primarily as leak-free hold-out RMSE, complemented where

feasible by approximate information-criteria-based comparisons (WAIC). All evaluations use leak-free standardization and are repeated for 4, 6, and 8 bins to assess robustness to binning resolution.

2.5 Recovery and falsification analyses

We ran a set of diagnostic checks on the pooled profiling procedure. First, we generated synthetic interference curves across a wide range of true angles and re-estimated the pooled posterior with the same pipeline (100 replicates per angle). Second, we generated quadratic null curves to test whether smooth non-interference structure could still produce a concentrated angular posterior. Third, we permuted the within-curve x ordering to remove temporal structure while preserving marginal latency values (500 replicates). Fourth, we bootstrap-resampled participants to assess the stability of the pooled MAP (100 resamples of $N = 30,000$). Fifth, we examined recovery as a function of sample size N . Finally, we compared posterior density, RSS, hold-out RMSE, and log score on a common θ grid. These checks are diagnostic. They constrain interpretation, but they do not by themselves establish mechanism.

To test whether the recovered geometry carries leverage beyond the same pooled curves used to estimate it, we added three external-validity analyses.

D-like congruency quintiles. Within the Gender–Science corpus we computed a conventional D-like congruency score for each participant, defined as the difference between mean incongruent and congruent latencies divided by the participant’s SD over critical-block latencies. Congruent and incongruent labels were derived from the raw `block_pairing_definition` field rather than from block numbers alone. We then reprofiled θ separately within D-like quintiles.

Matched public Sexuality IAT. We analyzed matched 2019 public raw logs from a Sexuality IAT in the same Project Implicit archive family, using the same critical blocks,

binning resolutions, and leak-free hold-out pipeline as in the main analysis. The Sexuality archive contained six split `iat*.txt` members; one file (`iat2.0003.txt`) was malformed and yielded no readable trial-level columns, so the matched-domain analysis used the remaining five raw logs. This comparison keeps the broad online IAT architecture, raw-log format, and preprocessing scheme close to the Gender–Science analysis while changing construct domain. The Sexuality comparison is more closely matched architecturally than the life-satisfaction task, but it still does not isolate construct content from all design features because stimuli and participant composition also differ.

Independent life-satisfaction IAT audit. We analyzed an independent 2024 life-satisfaction IAT collected in a five-phase Testable protocol (10 true/false practice trials, 10 self/not-self practice trials, 20 congruent combined trials, 10 reversed self/not-self practice trials, and 20 incongruent combined trials). The raw archive contained 187 per-participant CSV files. We kept the first attempt for repeated participant names, excluded participants with more than 10% of raw attempts faster than 300 ms, and collapsed repeated correction loops back to the original item by summing latency across attempts. This yielded 180 valid participants. Because the task uses sentence-like materials and a different five-phase architecture, it was treated as a stimulus-order sensitivity audit rather than as a matched construct contrast. We profiled the critical combined phases using both item identifiers and actual display order as the within-block axis, and we also examined stimulus-family balance between logical anchors and life-satisfaction statements.

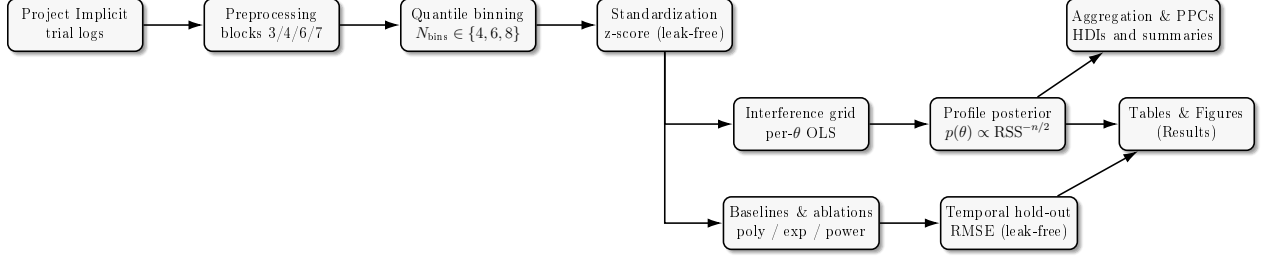


Figure 1: Processing pipeline for the profiled-posterior and hold-out analyses.

2.6 Pipeline diagram

3 Results

3.1 Posterior over θ

On $N = 141,329$ participants in the six-bin analysis, the profile posterior is sharply concentrated (Fig. 2). The posterior mean is 17.29° , the 94% HDI is $[16.50^\circ, 18.25^\circ]$, and the MAP is 17.25° . Aggregate posterior-predictive curves track the observed mean curves closely, with residual $\hat{\sigma} \approx 0.63$ (Fig. 3). Across binning resolutions, the posterior remains concentrated in the low-angle regime, although the mode shifts from 22.75° at 4 bins to 17.25° at 6 bins and 15.50° at 8 bins (Table 7).

Bootstrap resampling shows that the pooled estimate is highly stable at large N . Across 100 resamples of 30,000 participants per replicate, the pooled MAP was effectively constant: mean 17.00° , SD 0.00° , IQR 0.00° , with a 95% interval of $[17.0^\circ, 17.0^\circ]$ (Fig. 5).

The inferential optimum and the predictive optimum are not the same. On a common θ grid, the posterior peaks at a low interior angle, whereas hold-out RMSE is minimized at the boundary $\theta = 0^\circ$ for 4, 6, and 8 bins (Fig. 6). The pooled posterior therefore identifies a stable inferential solution, not the parameter value that minimizes short-horizon prediction error.

That divergence is a principled model-objective mismatch rather than a contradiction. Hold-out RMSE is a *local predictive criterion*: it rewards short-horizon interpolation of the

last bins after participant-wise refitting. The profiled interference angle targets a *global structural parameter*: the pooled geometric relation that best organizes contextual overlap across all curves. Figure 6 makes the mismatch explicit. The criterion that best recovers the stable low-angle posterior need not minimize one-step-ahead bin prediction.

3.2 External leverage and stimulus-order audit

The recovered angle is not merely a relabeling of conventional IAT effect size. In the Gender–Science corpus, D-like congruency scores were recomputed from the raw pairing labels rather than from block numbers alone. Mean D-like scores varied across participant quintiles from -0.320 in the lowest quintile to 0.825 in the highest. Yet the subgroup posteriors remained tightly clustered in the same low-angle regime: quintile MAPs ranged only from 17.00° to 17.75° , with HDIs spanning roughly 15° – 20° (Table 4). The angular summary therefore persists even when the corrected congruency metric changes markedly.

The matched public-domain comparison addresses a more stringent version of the same question. Using year-matched 2019 raw archives from the same Project Implicit family, the Sexuality IAT yields another sharply concentrated low-angle posterior rather than an orthogonal shift. In the six-bin analysis, the Sexuality posterior mean is 19.12° , with SD 0.53° , MAP 19.0° , and 94% HDI $[18.25^\circ, 20.0^\circ]$ across 207,366 pooled curves (Table 1; Fig. 4). Relative to the matched 2019 Gender–Science archive, the posterior contrast Sexuality – Gender–Science has mean 1.82° , 94% HDI $[0.50^\circ, 3.00^\circ]$, and posterior probability 0.991 of being greater than zero. Thus the shift is real but modest: both matched public domains remain in the same low-angle regime.

Across binning resolutions, the matched public domains show parallel behavior. In the 2019 Gender–Science subset, MAP estimates move from 22.75° at 4 bins to 17.25° at 6 bins and 15.50° at 8 bins. In Sexuality, the corresponding values are 25.50° , 19.00° , and 17.00° . The absolute levels differ slightly, but the same downward shift with finer binning and the same low-angle concentration are preserved in both tasks. The matched-domain evidence

therefore suggests that low geometric separation is not unique to the Gender–Science IAT.

Response-time descriptives were then recomputed with congruent and incongruent stages defined from the actual raw-log pairing labels (Table 2). This correction resolves the earlier apparent inconsistency in the Sexuality task: after using `block_pairing_definition`, incongruent trials are slower than congruent trials in all three public social-category domains. Gender–Science increases from a mean of 923.92 ms in congruent trials to 1081.54 ms in incongruent trials, Sexuality from 1032.90 ms to 1134.20 ms, and Age from 831.99 ms to 1058.41 ms.

The independent life-satisfaction IAT initially appeared to yield a qualitatively different geometry. When the critical combined phases were profiled using item identifiers as the within-block axis, the posterior concentrated at the high-angle boundary (mean 177.48° , MAP 180.0° , 94% HDI $[172.75^\circ, 180.0^\circ]$). A stimulus-order audit showed that this estimate should not be interpreted as evidence for a true near-orthogonal task regime. Each critical stimulus appeared exactly 180 times in each phase, so the issue was not unequal item counts. Instead, item identifiers 1–10 were logical true/false anchors and item identifiers 11–20 were life-satisfaction statements; the two stimulus families differed strongly in latency and error rate. When actual display order was used as the within-block axis, the MAP dropped to 17.75° ; after centering within stimulus family, the MAP was 18.50° (Table 3). The life-satisfaction task is therefore retained as a cautionary sensitivity audit rather than as evidence for a distinct near-orthogonal regime.

Table 1: Matched 2019 public-domain pooled θ summaries.

Task	Year	N	Theta mean	Theta SD	Theta MAP	HDI low	HDI high
Gender-Science	2019	138955	17.294	0.488	17.250	16.500	18.250
Age	2019	156122	18.476	0.522	18.500	17.500	19.250
Sexuality	2019	207366	19.119	0.532	19.000	18.250	20.000

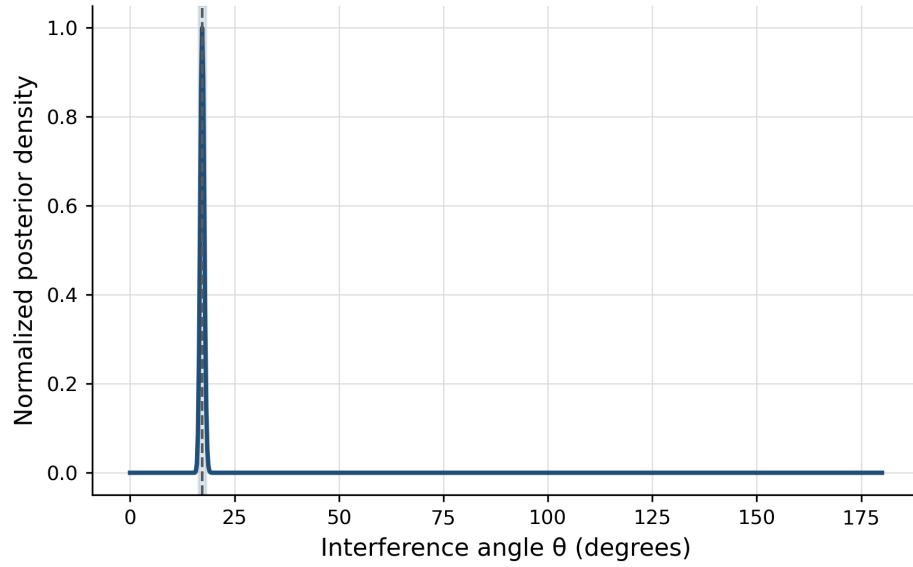


Figure 2: Posterior over θ from grid-based OLS profiling on the full sample. Shaded band: 94% HDI.

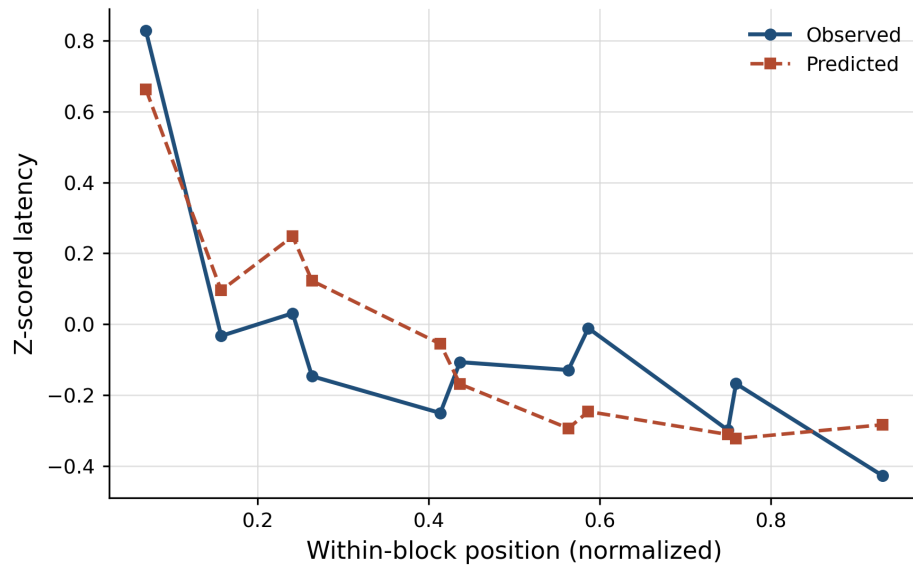


Figure 3: Observed vs. model-predicted aggregate curves at $\theta_{\text{MAP}} = 17.25^\circ$. Points: observed binned means; dashed: model prediction.

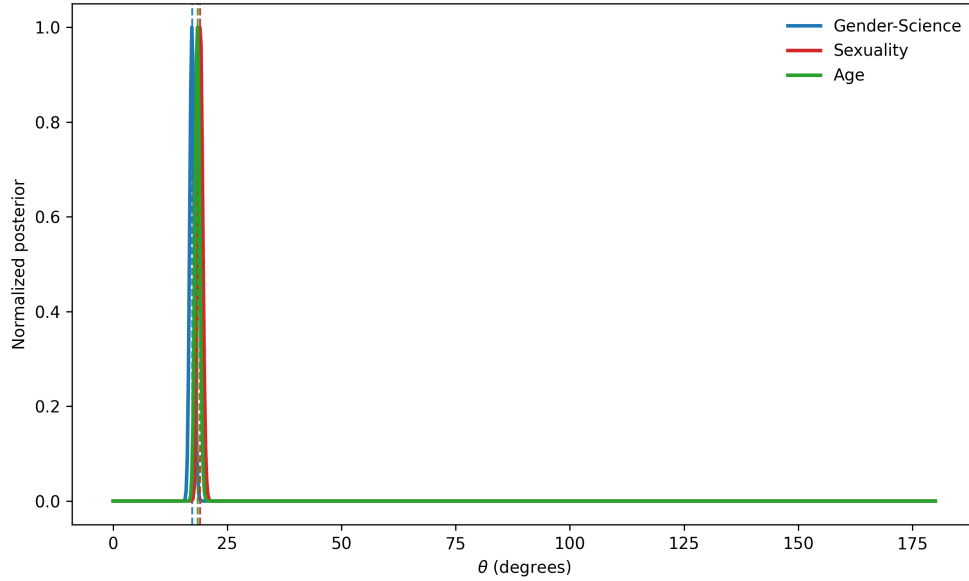


Figure 4: Matched public 2019 raw-domain comparison from the same Project Implicit archive family. Both Gender–Science and Sexuality produce sharply concentrated low-angle posteriors. Sexuality shifts upward slightly, but remains in the same low-angle regime as Gender–Science.

Table 2: Response-time descriptives by corrected task stage. Congruent and incongruent stages are defined from the raw pairing labels rather than from block numbers alone. Values are in milliseconds.

Task	Stage	Trials	Mean RT	Median RT	SD	IQR
Age	Congruent	8,978,250	831.99	680.00	555.28	303.00
Age	Incongruent	8,928,856	1058.41	828.00	757.48	549.00
Gender–Science	Congruent	8,006,219	923.92	750.00	599.31	405.00
Gender–Science	Incongruent	7,972,388	1081.54	876.00	723.23	543.00
Life-satisfaction IAT	Congruent	3,600	2515.39	2001.00	2331.73	1524.75
Life-satisfaction IAT	Incongruent	3,600	3077.98	2376.00	2863.49	1881.25
Sexuality	Congruent	11,897,059	1032.90	830.00	694.40	474.00
Sexuality	Incongruent	11,884,564	1134.20	911.00	776.87	566.00

Table 3: Life-satisfaction IAT stimulus-order sensitivity audit for the pooled angular profile.

Variant	Mean θ	SD	MAP	HDI low	HDI high
Item-ID axis, all critical items	177.48	2.68	180.00	172.75	180.00
Display-order axis, all critical items	28.49	13.68	17.75	8.50	53.75
Item-ID axis after within-family centering	42.02	25.05	18.50	7.25	87.75
Display-order axis, logical anchors only	25.34	11.85	16.25	8.00	47.25
Display-order axis, life-satisfaction items only	35.44	19.79	17.50	7.50	71.75

Table 4: Gender–Science pooled θ by D-like congruency-score quintile.

Quintile	N	Mean D-like	Mean θ	SD θ	MAP θ	HDI low	HDI high
1	26,743	-0.320	17.982	1.247	17.750	15.750	20.250
2	26,743	0.079	17.178	1.066	17.000	15.250	19.000
3	26,742	0.306	17.139	1.052	17.000	15.250	19.000
4	26,743	0.516	17.139	1.070	17.000	15.250	19.000
5	26,743	0.825	17.716	1.203	17.500	15.500	20.000

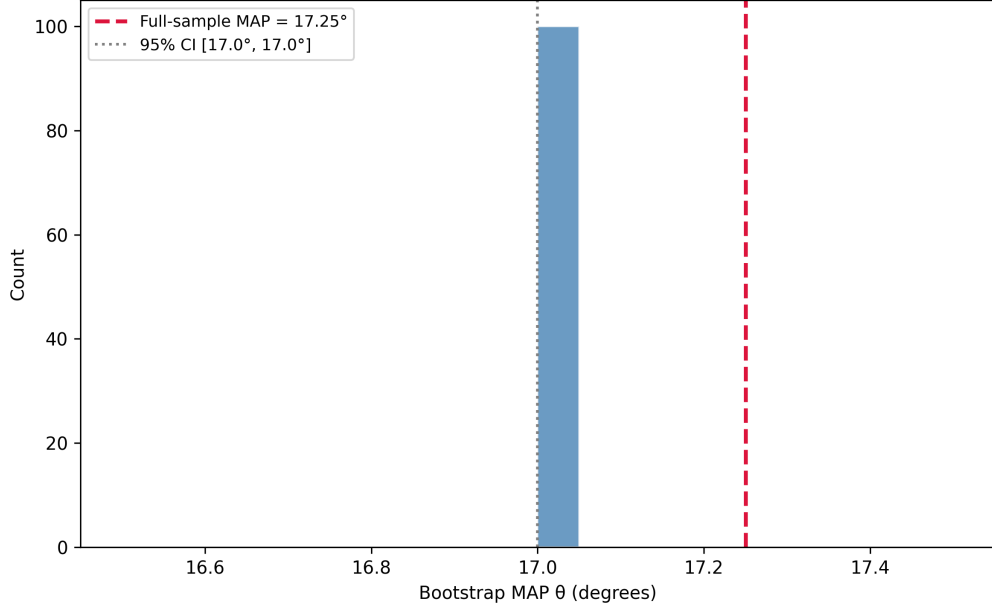


Figure 5: Bootstrap stability of the pooled θ estimate. Each replicate resamples 30,000 participants and recomputes the pooled MAP. The bootstrap distribution is effectively degenerate around the empirical solution, indicating extremely high stability of the population-level estimate at large N .

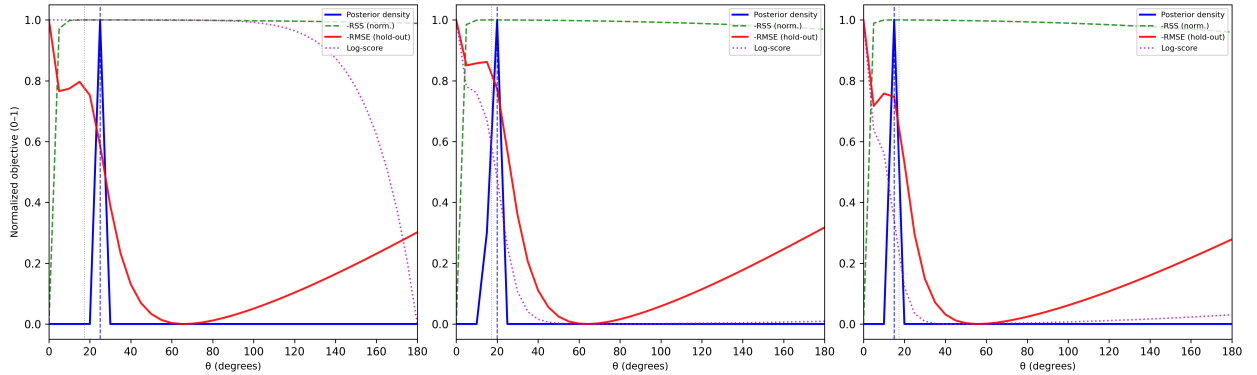


Figure 6: Population-level objectives evaluated on the same θ grid for 4, 6, and 8 bins. The posterior density peaks in the low-angle interior, whereas the hold-out RMSE optimum occurs at $\theta = 0^\circ$ in all three binning resolutions.

Table 5: Posterior summary for θ in the main six-bin analysis.

Parameter	Mean	SD	MAP	94% HDI low	94% HDI high
θ (deg)	17.29	0.48	17.25	16.50	18.25

Table 6: Overall pooled fit quality at θ_{MAP} .

Curves	Binned observations	RSS(θ_{MAP})	$\hat{\sigma}$	Posterior peak
141329	847974	336141.06	0.630	0.2068

3.3 Predictive validation

In the six-bin Gender–Science analysis, the quadratic baseline is essentially tied with the full interference model on participant-level temporal hold-out RMSE (21.924 vs. 21.919), while some other additive or boundary cases do slightly better on this local criterion (Fig. 7; Table 8). Approximate WAIC on the pooled fit also favors the quadratic baseline over the interference model, while the exponential and power-law baselines perform much worse (Table 10). The same broad pattern appears across bin counts (Table 9). Read narrowly, these results say that smooth additive approximations can be competitive for short-horizon prediction on the pooled public corpus. They do *not* show that the additive models recover the same structure as the geometric fit.

The matched public Sexuality IAT reproduces the same local-prediction pattern. In that domain, the full interference model and the quadratic baseline are again nearly indistinguishable on temporal hold-out RMSE (25.015 vs. 25.019), while cosine-only, sine-only, exponential, and power-law variants are only marginally lower (24.997–25.000). The near-tie between geometric and additive prediction is therefore not specific to

Table 7: Pooled profile posterior summary by binning resolution.

Bins	Mean θ	SD	MAP	HDI low	HDI high	Posterior entropy	$\hat{\sigma}$ at MAP
4	22.901	0.623	22.750	21.750	24.000	2.330	0.443
6	17.292	0.483	17.250	16.500	18.250	2.077	0.630
8	15.566	0.423	15.500	14.750	16.250	1.944	0.697

Gender–Science; it appears in a second public IAT processed through the same raw-log pipeline.

The independent life-satisfaction task is not used as a positive predictive contrast in the revised interpretation. The stimulus-order audit showed that its apparent high-angle solution depends on using item identifiers as the within-block axis. Predictive comparisons tied to that same axis are therefore treated as preprocessing diagnostics, not as evidence that the task occupies a genuine near-orthogonal contextual regime.

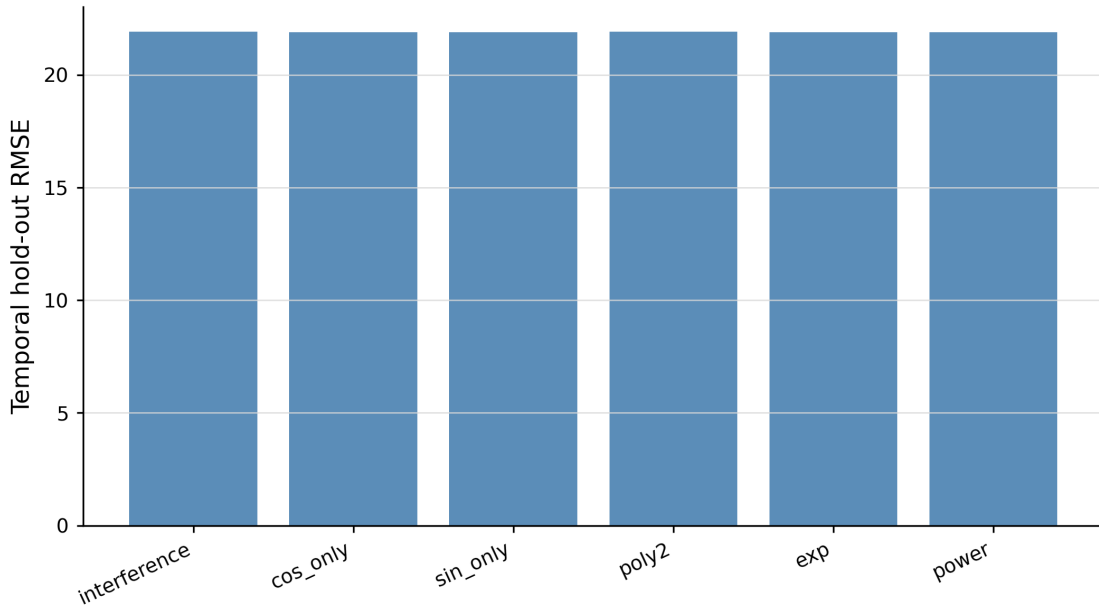


Figure 7: Temporal hold-out RMSE for the six-bin analysis. Differences between models are small in absolute magnitude, but the full interference model does not achieve the lowest hold-out error.

Table 8: Leak-free temporal hold-out comparison for the six-bin analysis (lower RMSE is better).

Model	Best parameter	RMSE _{test}	Mean log score _{test}	σ_{train}	Boundary hit
Interference	124.20	21.919	-21524.695	0.1056	No
Cosine only	180.00	21.899	-7212.871	0.1823	Yes
Sine only	180.00	21.897	-9632.215	0.1578	Yes
Polynomial (2 nd order)	—	21.924	-21602.217	0.1055	No
Exponential	10.10	21.896	-9905.413	0.1556	No
Power law	0.05	21.897	-9259.726	0.1609	Yes

Table 9: Interference-model hold-out robustness across numbers of bins per participant.

Bins	Selected θ (deg)	RMSE _{test}	σ_{train}	Boundary hit
4	180.00	19.646	0.0000	Yes
6	124.20	21.919	0.1056	No
8	76.60	26.165	0.1703	No

Table 10: Approximate WAIC comparison on the pooled full-sample fit. Higher ELPD-WAIC is better.

Model	ELPD-WAIC	p_{WAIC}	SE	$\Delta\text{ELPD-WAIC}$
Polynomial (2 nd order)	-810739.770	0.000	636.685	0.000
Interference	-811813.911	0.012	632.836	-1074.141
Exponential	-918121.539	1.514	660.983	-107381.769
Power law	-927029.858	0.000	607.930	-116290.088

3.4 Residual structure and held-out distributional fit

Near-tied RMSE does not imply that the quadratic baseline captures the same empirical structure. On block-curve hold-out data, the interference and quadratic fits are almost indistinguishable by RMSE alone (57.218 vs. 57.235), yet their residual maps are different (Fig. 8; Table 11). The quadratic model leaves stronger negative drift in the last held-out bin, especially in the higher-angle blocks 4 and 7. Its mean absolute residual in the final held-out bin of those higher-angle blocks is 0.476, versus 0.381 for the interference fit. The magnitude of quadratic residuals also aligns more strongly with the blockwise pooled angle (correlation 0.240 vs. 0.130). The additive fit therefore leaves structured error that tracks contextual geometry.

Going beyond RMSE to the *distributional shape* of the held-out responses strengthens the same conclusion. Across the eight held-out block/bin cells, the interference fit yields lower Wasserstein distance, lower quantile error, and lower interquartile-range error in every cell (means 1.022 vs. 1.323, 0.956 vs. 1.283, and 1.524 vs. 2.070 respectively; Fig. 9; Table 11). The additive model can therefore match local pointwise error while still misrepresenting the global shape of the held-out response distribution.

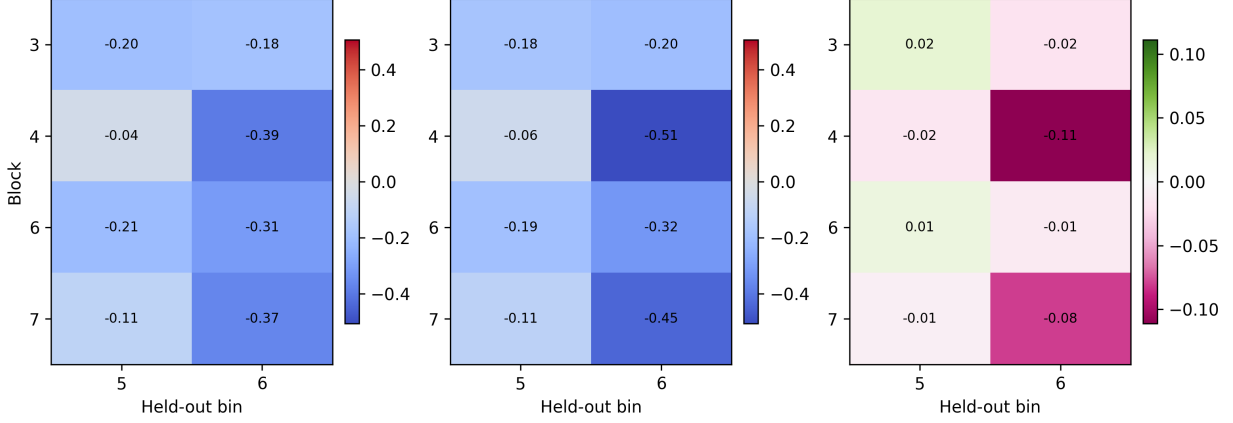


Figure 8: Held-out residual structure on block curves. Left: interference-model residual means by block and held-out bin. Middle: quadratic-model residual means. Right: quadratic minus interference residual means. The quadratic baseline leaves stronger late-bin negative drift, especially in higher-angle blocks.

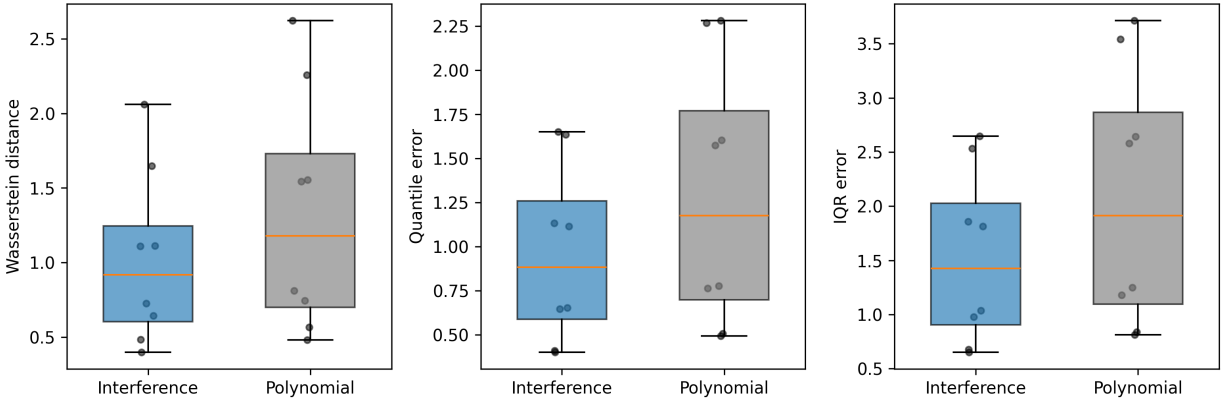


Figure 9: Held-out distributional-fit diagnostics across the eight critical block/bin cells. Lower values are better. In every cell, the interference model yields lower Wasserstein distance, quantile error, and interquartile-range error than the quadratic baseline.

Table 11: Model-objective-mismatch diagnostics on block-curve hold-out data. Lower is better except for the residual- θ correlation, where values closer to zero indicate less geometry-aligned structured error.

Metric	Interference	Polynomial
Block-curve RMSE	57.218	57.235
Mean abs. held-out cell residual	0.226	0.252
High-angle final-bin abs. residual	0.381	0.476
Corr(abs. residual, block θ)	0.130	0.240
Mean Wasserstein distance	1.022	1.323
Mean quantile error	0.956	1.283
Mean IQR error	1.524	2.070

3.5 Recovery and falsification analyses

Recovery is strongly asymmetric across the parameter space (Fig. 10; Table 12). For low true angles, recovered MAPs are biased upward: mean recovered MAPs are 41.743° for a true 5° , 39.763° for 10° , 43.173° for 17.25° , and 41.613° for 20° . In this range, 94% HDI coverage is zero or near zero. Coverage improves at 25° and is close to nominal from 40° upward. Under sparse binning, the pooled procedure therefore identifies moderate and high angles far more reliably than low ones.

Bootstrap and sample-size analyses explain why the empirical estimate is nevertheless stable. Bootstrap resampling reproduces the pooled angle almost exactly (Fig. 5). In the recovery-versus- N analysis, small and moderate samples remain biased upward and often miss nominal coverage, whereas the empirical dataset at $N = 141,329$ lies in a regime where the recovered MAP coincides with the observed estimate and the HDI covers the target angle (Fig. 11).

The null diagnostics are mixed rather than uniformly favorable (Fig. 12; Table 13). Relative to the quadratic null family, the empirical posterior is more sharply concentrated and centered at a lower angle. The within-curve permutation control is less decisive: some permuted datasets still yield concentrated, boundary-seeking posteriors, so posterior concentration alone is not a sufficient falsification diagnostic. The null analyses therefore constrain interpretation, but they do not by themselves establish mechanism.

3.6 Block-dependent phase structure

To examine whether the inferred interference angle reflects a stationary property of the task or an aggregate over heterogeneous contexts, we estimated block-specific pooled phase parameters. The resulting structure exhibits systematic variation across blocks (Fig. 13; Table 14). Block 3 is associated with a lower pooled angle, whereas blocks 4, 6, and 7 cluster at higher values. This pattern indicates that contextual overlap is not constant over the course of the IAT and suggests that the aggregate interference angle recovered at the

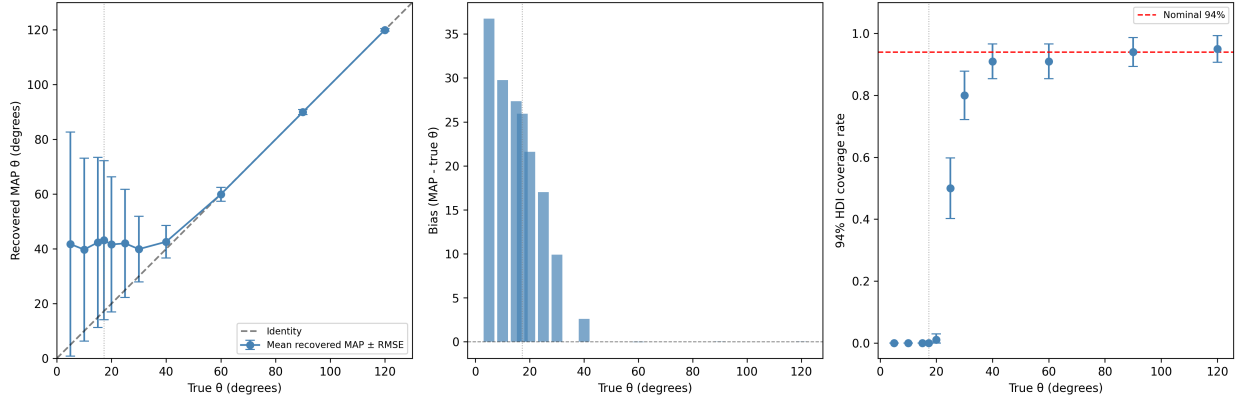


Figure 10: Extended simulation-based recovery of pooled θ . Left: identity plot with RMSE bars. Middle: bias as a function of the true generating angle. Right: 94% HDI coverage as a function of the true generating angle. Recovery is poor in the low-angle regime and becomes near-nominal only at moderate-to-high angles.

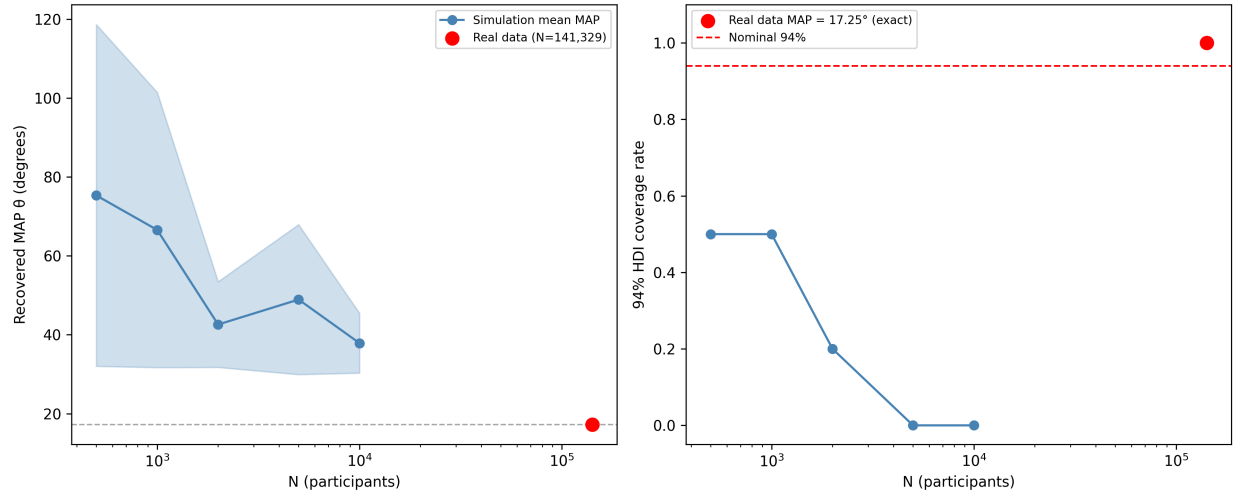


Figure 11: Population-level recovery as a function of sample size for the low-angle regime around the empirical solution. Simulated recovery remains unstable at modest N , whereas the empirical dataset at $N = 141,329$ lies in the regime where the recovered MAP coincides with the observed solution and the HDI covers the target.

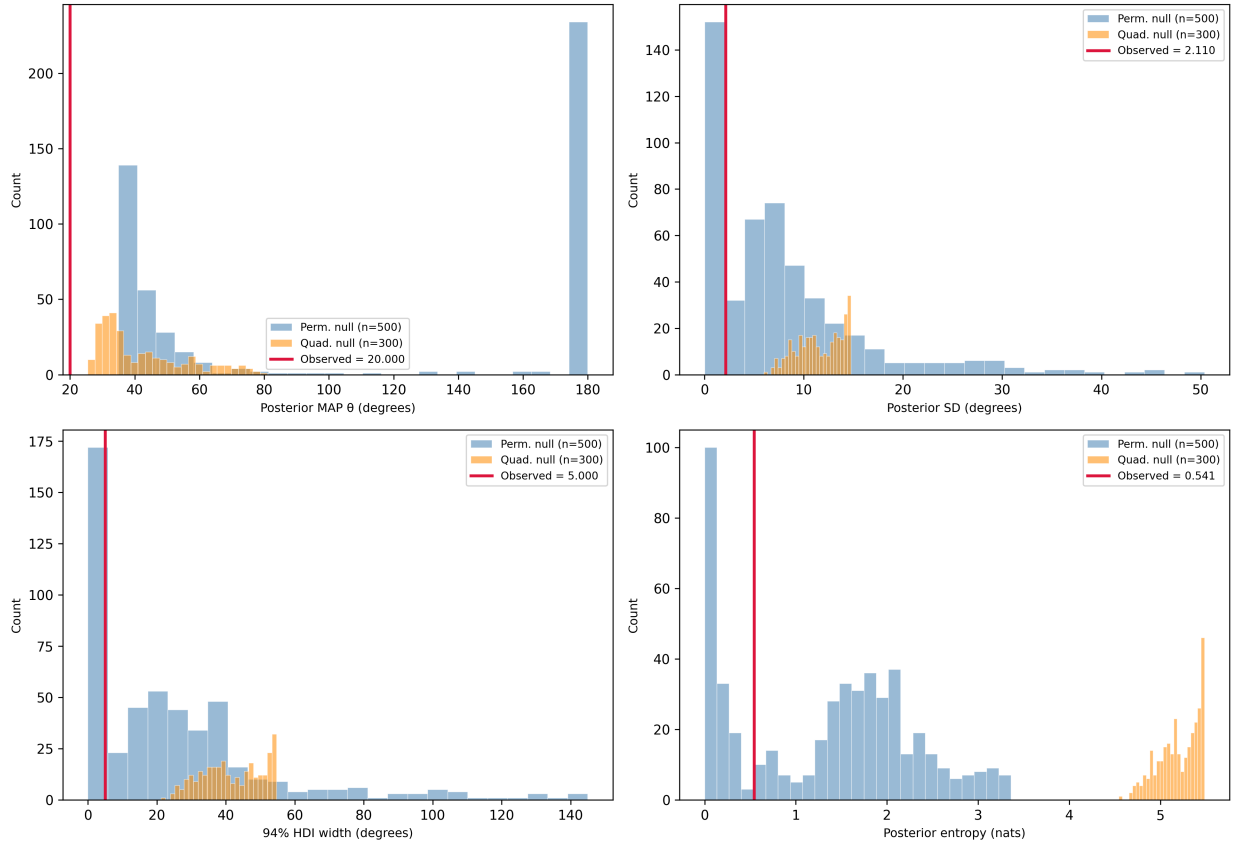


Figure 12: Null posterior diagnostics versus the observed fit. The empirical posterior differs clearly from the quadratic additive null, whereas the within-curve permutation control remains a weaker and non-decisive falsification benchmark.

Table 12: Expanded simulation-based recovery over the full 0° – 180° profiling grid (100 replicates per angle, $N = 5,000$ curves each).

True θ	Replicates	Mean recovered MAP	Bias	RMSE	Median posterior SD	94% HDI coverage	Med.
5.000	100	41.743	36.743	40.930	14.394	0.000	
10.000	100	39.763	29.763	33.412	12.658	0.000	
15.000	100	42.363	27.363	31.090	12.451	0.000	
17.250	100	43.173	25.923	29.044	11.810	0.000	
20.000	100	41.613	21.613	24.666	10.752	0.010	
25.000	100	42.030	17.030	19.739	8.620	0.500	
30.000	100	39.915	9.915	11.987	6.995	0.800	
40.000	100	42.610	2.610	5.964	5.172	0.910	
60.000	100	59.948	-0.053	2.524	2.356	0.910	
90.000	100	89.988	-0.013	0.962	0.885	0.940	
120.000	100	119.990	-0.010	0.494	0.438	0.950	

Table 13: Summary of posterior-shape diagnostics under the two null families.

Null type	Replicates	MAP mean	MAP SD	Mean posterior SD	Mean HDI width	Mean entropy
Permutation	500	109.650	67.790	7.716	24.120	1.317
Quadratic	300	42.250	13.590	11.591	41.700	5.200

population level represents a mixture over block-specific geometries rather than a single stationary parameter.

Figure 13 makes the additive comparison sharper. The pooled angular summaries move substantially across blocks, from 21° in block 3 to roughly 26° – 29° in blocks 4, 6, and 7. By contrast, the blockwise quadratic coefficients cluster in almost the same direction in the (d_1, d_2) plane (coefficient angles 145° – 148°) and vary mostly in norm rather than interpretable orientation. The polynomial model can only absorb block-dependent phase variation by refitting nuisance coefficients separately within each block; those coefficients do not map onto a single geometric dimension with substantive interpretation. θ , by contrast, places the blocks on a common contextual-rotation scale. This is exactly the kind of low-dimensional structure that local RMSE does not reward.

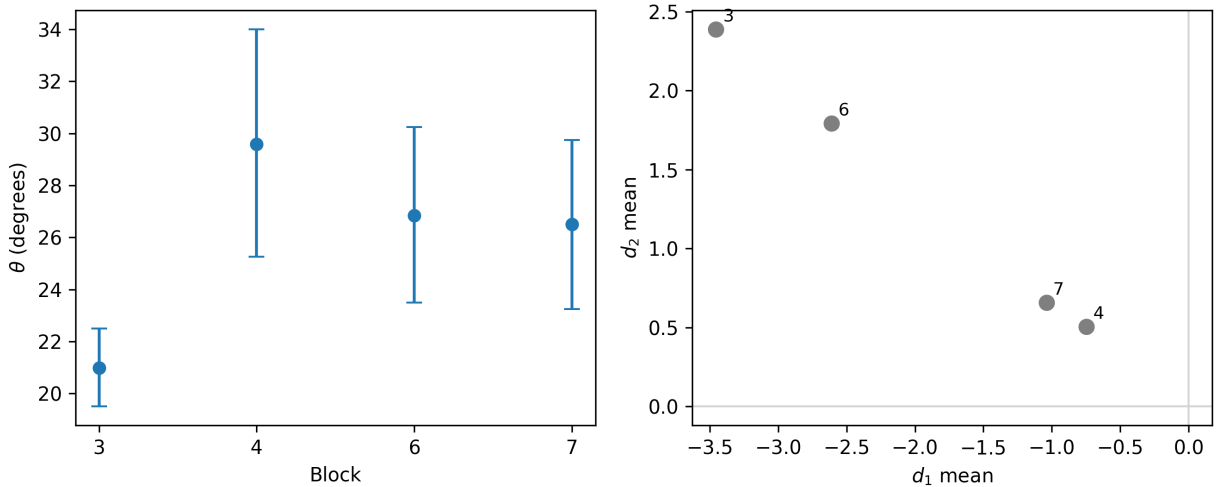


Figure 13: Blockwise geometric versus additive summaries. Left: block-specific pooled θ estimates with profiled HDI bounds. Right: block-specific quadratic coefficients in the (d_1, d_2) plane. The angular summaries move systematically across blocks, whereas the quadratic coefficients largely preserve direction and vary mostly in norm.

Table 14: Block-specific pooled θ summaries.

Block	Curves	Mean θ	SD	HDI low	HDI high	MAP
3	141258	20.977	0.828	19.500	22.500	21.000
4	140027	29.580	2.388	25.250	34.000	29.000
6	136487	26.836	1.807	23.500	30.250	26.500
7	135700	26.503	1.751	23.250	29.750	26.250

Table 15: Model comparison on block-curve hold-out data.

Model	Best parameter(s)	RMSE _{test}	Mean log score _{test}	Boundary hit
Interference (global θ on block curves)	180.0	57.218	-32778.299	Yes
Interference (block-specific θ)	3: 180.0; 4: 136.8; 6: 180.0; 7: 137.5	57.220	-26240.180	Yes
Polynomial (block curves)	—	57.235	-32781.817	No

Table 16: Per-block hold-out details for the block-specific θ selection procedure.

Block	Selected training θ	RMSE _{test}	Mean log score _{test}	Boundary hit
3	180.000	111.773	-99475.387	Yes
4	136.800	9.893	-1438.477	No
6	180.000	14.863	-1592.495	Yes
7	137.500	5.099	-388.577	No

3.7 Quantifying block-level phase variation

Block-specific phase estimates exhibit systematic dispersion across critical blocks. Wrapped phase summaries are reported in Table 17. Taken together with the block-specific pooled-angle differences above, these results indicate that the pooled contextual overlap angle is best interpreted as an aggregate over heterogeneous block-specific geometries rather than as a stationary phase parameter.

Table 17: Wrapped phase summaries for each critical block at the pooled global θ_{MAP} .

Block	Curves	Circular mean (deg)	Circular variance	Global θ_{MAP}
3	141258	-159.793	0.660	17.250
4	140027	-151.569	0.952	17.250
6	136487	-158.427	0.753	17.250
7	135700	-151.705	0.922	17.250

4 Discussion

The main result is a stable low-angle interference solution, $\theta \approx 17^\circ$, for the full Gender–Science corpus. That solution is sharply concentrated in the empirical data, stable across bootstrap resampling, and still visible across binning resolutions. It also remains essentially unchanged across corrected D-like strata. A matched 2019 public Sexuality IAT from the same Project Implicit archive family lands in the same broad low-angle regime, with a modest upward shift to about 19° . The independent life-satisfaction task initially appeared to shift to a near-orthogonal solution near 180° , but sensitivity checks showed that this was an item-axis artifact: using actual display order or centering stimulus family returned low-angle MAPs. A small and stable θ should not be read as evidence that the categories in the Gender–Science IAT are theoretically identical. In the present model, θ summarizes the separation of task contexts in aggregated latency dynamics. Thus $\theta \approx 17^\circ$ indicates low behavioral separation between the congruent and incongruent task regimes, not semantic identity of the underlying constructs.

This distinction clarifies why the result is theoretically informative rather than contradictory. The Gender–Science IAT is built around nominally incongruent pairings, yet the pooled trajectories do not behave like strongly incompatible contexts. In the Gender–Science corpus, θ barely moves across participant D-like strata, but it does move across task phases: from the full-sample value near 17° to block-specific values in the 21° – 29° range. The most stable IAT structure in these analyses is therefore task-geometric rather than participant-specific. What often looks like a single global learning or trait signal is better described as a mixture over differently rotated contexts.

The predictive results should therefore be interpreted through an explicit model–objective distinction. RMSE is a *local predictive criterion*; it rewards short-horizon interpolation of the last held-out bins after each curve is refit. The profiled interference angle is a *global structural parameter*; it is intended to summarize contextual overlap across the pooled task geometry. These objectives are not aligned, as Fig. 6 shows directly. In the public corpus, a quadratic approximation can be competitive on local RMSE while still leaving stronger residual drift and worse held-out distributional fit. The right conclusion is therefore not that the interference model “fails” on prediction, but that predictive and structural targets reward different aspects of the data.

What produces the low separation in the Gender–Science task remains unresolved. One class of explanation is structural: shared response mappings, stimulus familiarity, lexical length, stimulus type, and other design properties may compress the geometry of the task. Recent IAT work supports this possibility by showing that measurement properties depend on stimulus validation and response modality (Hogenboom et al., 2024; Bar-Anan and Hershman, 2024), and that predictive validity differs across indirect-measure architectures (Axt et al., 2024). A second class of explanation is construct-population: if gender–science associations are comparatively less divergent in the sampled population, then even a well-constructed IAT may yield low geometric separation. The present data do not cleanly separate these explanations.

The new matched Sexuality analysis helps answer the sharper theoretical question: is low geometric separation mainly a property of this test architecture, or does it arise because the Gender–Science constructs are only weakly divergent in the sampled population? The present evidence points to a joint explanation with substantial task-architecture influence. Sexuality is ordinarily treated as a more consolidated social-category contrast than gender–science, yet under a closely matched public IAT architecture it still produces a low-angle posterior. The shift relative to the matched 2019 Gender–Science subset is statistically credible but modest (posterior mean difference 1.82° , 94% HDI $[0.50^\circ, 3.00^\circ]$, $P(\theta_{\text{Sexuality}} > \theta_{\text{Gender–Science}}) = 0.991$). Stronger construct content alone therefore does not push the system toward orthogonality. At the same time, the nonzero shift cautions against treating θ as a pure task artifact. The safest interpretation is that θ reflects a joint property of task architecture and population-level construct separation, with architecture constraining the range strongly enough that even a more consolidated public-domain contrast remains low-angle.

The life-satisfaction audit speaks to the same question in a different way. It shows that the profiling procedure can be highly sensitive to how the within-block axis is encoded when stimulus families are ordered by item identifier. In that task, logical true/false anchors occupy item identifiers 1–10 and life-satisfaction statements occupy item identifiers 11–20. The two families differ strongly in latency and error rates, so item ID becomes partly a stimulus-family axis. The near-orthogonal result therefore should not be interpreted as a substantive task contrast. Instead, it provides a useful diagnostic warning: cross-task comparisons require stimulus-level and order-level audits before differences in θ are interpreted psychologically.

The recovery analyses still narrow what can be claimed. Under sparse binning, low true angles are not well recovered in small or moderate synthetic samples; they are biased upward and show poor interval coverage. The empirical low-angle estimate becomes stable only under very large aggregation. That makes θ interpretable as a population-level descriptor of task geometry, not as a reliable participant-level trait.

The null analyses sharpen the same point, but asymmetrically. Posterior concentration alone is not enough to support a mechanistic interpretation. The empirical solution differs clearly from the quadratic additive null, but the within-curve permutation null is not cleanly defeated. The safest conclusion is therefore limited: the observed solution is not well described as a simple quadratic artifact, yet the null analyses do not on their own rule out all alternative sources of posterior concentration.

The strongest claim is therefore not predictive superiority. It is that additive baselines can win or tie on short-horizon RMSE while still leaving evidence of structured, low-dimensional geometry in the data. The interference model recovers that geometry in a way that is stable and identifiable at the population level, while the life-satisfaction audit shows that cross-task contrasts can be distorted by stimulus-order artifacts. In the public corpus, the angular summary survives large differences in a corrected conventional congruency metric and replicates across a matched public Sexuality domain. The model does not establish that contextual interference is always the best predictive model of trial-aggregated latency curves, and it does not justify strong claims about individual latent structure. But it does constrain how IAT structure should be interpreted: the strongest stable signal in these data is geometric and contextual, provided that stimulus and ordering confounds are checked explicitly.

Future directions. Three follow-up strategies could discriminate task-structure from construct-population explanations more directly. First, one can manipulate stimulus sets within the same IAT architecture, for example by swapping words, images, or culturally localized exemplars; systematic shifts in θ under matched procedures would indicate design sensitivity (Hogenboom et al., 2024). Second, one can test country-level variation with archives that retain national identifiers, such as Project Implicit:International (Charlesworth et al., 2023); if θ varies with cultural context, that would implicate population-level stereotype structure rather than a fixed property of the task. Third, the same geometric

pipeline should now be extended beyond Sexuality to additional matched public domains, especially race, to determine whether the low-angle regime generalizes across multiple consolidated social contrasts or whether some domains truly approach orthogonality under the same online IAT architecture.

Limitations. The data are observational and self-selected. Residuals are modeled as Gaussian. Participant-specific coefficients are treated as fixed effects. Participant-level interference angles remain weakly identified under sparse binning. Stronger claims about individual variation would require richer within-participant time series or a trial-level hierarchical model. The present trial-level corpus is a large online Project Implicit sample, but the analyzed logs contain task and session fields rather than country identifiers, so we cannot determine the national composition of the sample or test cross-cultural moderation within this corpus. In the matched Sexuality archive, one split raw file was malformed and had to be excluded, although the remaining sample was still very large. Even with that new matched-domain comparison, the design remains observational rather than experimental: we still cannot cleanly decompose task structure, stimulus set, and population composition. The external life-satisfaction task is not psychometrically matched to the public IATs, and its original near-orthogonal estimate depends on using item identifiers as the within-block axis. It should therefore be treated as a preprocessing sensitivity case rather than as evidence for a distinct latent regime.

Data availability

The analyses use public Project Implicit raw logs for the Gender–Science and Sexuality IATs together with the external life-satisfaction task described in the Methods. Code and processed data will be made available upon publication.

Author contributions

All authors contributed to study conception, analysis, and manuscript preparation. All authors approved the final version.

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Conflicts of interest

The authors declare no competing interests.

Ethics statement

The study uses publicly available, anonymized data from the Project Implicit demo release. No additional ethical approval was required.

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